EX28 · Additional matrix algebra note

Building **A**, the alternative **Bx** $=$ **u** interpretation, and orthogonal matrices

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This supplementary note explains the same matrix problem through **two complementary operations**: constructing a Cartesian force from actuator coefficients, or computing actuator-axis components from a Cartesian force. No prior knowledge of coordinate-frame transformations is required.

1. Meaning of the direction vectors

Consider two actuators whose directions, expressed in the global Cartesian X, Y basis, are:

$$\mathbf{e}_1 = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}, \qquad \mathbf{e}_2 = \begin{pmatrix} -0.8 \\ 0.6 \end{pmatrix}.$$

Both vectors have **unit length**, and they are **perpendicular** to one another:

$$\|\mathbf{e}_1\|_2 = \sqrt{0.6^2 + 0.8^2} = 1, \qquad \|\mathbf{e}_2\|_2 = \sqrt{(-0.8)^2 + 0.6^2} = 1,$$
$$\mathbf{e}_1 \cdot \mathbf{e}_2 = 0.6(-0.8) + 0.8(0.6) = 0.$$

These are complete actuator directions, **not the global unit axes** $\hat{\imath} = (1, 0)$ and $\hat{\jmath} = (0, 1)$. Their entries are measured relative to the global axes.

2. First approach: combining directions into a force

The actuator force coefficients are $u_1 = 10$ N and $u_2 = 5$ N, collected in $\mathbf{u} = (10, 5)^T$ N. The resulting Cartesian force is:

$$\begin{aligned} \mathbf{x} = \mathbf{R} &= 10\mathbf{e}_1 + 5\mathbf{e}_2 \\ &= 10 \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix} + 5 \begin{pmatrix} -0.8 \\ 0.6 \end{pmatrix} = \begin{pmatrix} 6 \\ 8 \end{pmatrix} + \begin{pmatrix} -4 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 11 \end{pmatrix} \text{ N}. \end{aligned}$$

To express this **linear combination**, place each direction vector in a **column** of **A**:

$$\underbrace{\begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix}}_{\mathbf{A}=[\mathbf{e}_1\ \mathbf{e}_2]} \underbrace{\begin{pmatrix} 10 \\ 5 \end{pmatrix}}_{\mathbf{u}} = \underbrace{\begin{pmatrix} 2 \\ 11 \end{pmatrix}}_{\mathbf{x}=\mathbf{R}}. \qquad \mathbf{Au} = \mathbf{x}$$

Here the **rows** of **A** refer to Cartesian X and Y components, while the **columns** correspond to actuators 1 and 2. Multiplication by **u** adds both physical contributions.

3. Alternative approach: start from Bx $=$ u

Alternatively, build a matrix whose **columns describe the contribution of the global X and Y directions** to the actuator-axis components. In that case, define

$$\mathbf{B} = \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix}.$$

This is the transpose of the first matrix, $\mathbf{B} = \mathbf{A}^T$. Its first column $(0.6, -0.8)^T$ describes how one unit of Cartesian X contributes to projections along the two actuator directions. Its second column $(0.8, 0.6)^T$ does the same for Y.

The complete alternative formulation is:

$$\boxed{\mathbf{Bx} = \mathbf{u}}, \qquad \begin{pmatrix} 0.6 & 0.8 \\ -0.8 & 0.6 \end{pmatrix} \begin{pmatrix} x_x \\ x_y \end{pmatrix} = \begin{pmatrix} 10 \\ 5 \end{pmatrix}.$$

Here $\mathbf{x} = (x_x, x_y)^T$ is **the resultant force in Cartesian X, Y components**; it is not the actuator coefficient vector. The system projects **x** onto each actuator direction:

$$\mathbf{e}_1 \cdot \mathbf{x} = 0.6x_x + 0.8x_y = 10,$$
$$\mathbf{e}_2 \cdot \mathbf{x} = -0.8x_x + 0.6x_y = 5.$$

Solving from this formulation, before knowing x: since **B** is orthogonal, $\mathbf{B}^{-1} = \mathbf{B}^T$. Multiply by the inverse:

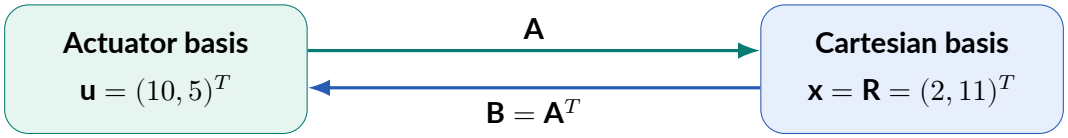
$$\mathbf{x} = \mathbf{B}^{-1}\mathbf{u} = \mathbf{B}^T\mathbf{u}$$
$$= \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix} \begin{pmatrix} 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 6-4 \\ 8+3 \end{pmatrix} = \begin{pmatrix} 2 \\ 11 \end{pmatrix} \text{ N}.$$

Checking the solution by substitution in $\mathbf{Bx} = \mathbf{u}$:

$$0.6(2) + 0.8(11) = 1.2 + 8.8 = 10,$$
$$-0.8(2) + 0.6(11) = -1.6 + 6.6 = 5.$$

Therefore, $\mathbf{Bx} = \mathbf{u}$ is **valid**: it represents the inverse operation to $\mathbf{Au} = \mathbf{x}$. One must not use **Bu** to obtain the resultant with the same variable meanings: $\mathbf{Bu} = (10, -5)^T$, not $(2, 11)^T$. **The difference is not an algebraic error; it is the meaning of the input and output coordinates.**

4. Direct and inverse transformation



The forward transformation combines actuator coefficients into a Cartesian force; the inverse projects the force onto the actuator basis.

5. Why is the transpose the inverse?

The columns of **A** form an **orthonormal basis**: each has unit length, and distinct columns have zero dot product. Thus **A** is **orthogonal** and satisfies:

$$\mathbf{A}^T\mathbf{A} = \mathbf{A}\mathbf{A}^T = \mathbf{I}, \qquad \mathbf{A}^{-1} = \mathbf{A}^T = \mathbf{B}.$$

Consequently, the same data can be recovered through either formulation:

$$\mathbf{Au} = \mathbf{x}, \qquad \mathbf{A}^T\mathbf{x} = \mathbf{u}, \qquad \mathbf{x} = \mathbf{B}^T\mathbf{u}.$$

Important limitation: unit vectors alone are not sufficient; they must also be mutually perpendicular. For a general matrix, $\mathbf{A}^T$ is **not** necessarily $\mathbf{A}^{-1}$, and plain dot-product projections do not recover the coefficient vector of a non-orthonormal basis.

6. Final interpretation and practical rules

Confusion can arise when treating the actuator directions themselves as the global X, Y axes. They are different objects: $\hat{\imath}$ and $\hat{\jmath}$ are the global axes; $\mathbf{e}_1$ and $\mathbf{e}_2$ are the actuator directions **expressed in that basis**.

- **To construct the Cartesian force**, place each direction vector in a **column of A**: $\mathbf{Au} = \mathbf{x}$.
- **To recover the actuator-basis coefficients**, place the directions as **rows of B**: $\mathbf{Bx} = \mathbf{u}$, where $\mathbf{B} = \mathbf{A}^T$ for an orthonormal basis.
- That is why each direction vector must be a **column of A, not a row**, *when* **A** *is defined to combine those directions with their coefficients.*
- This distinction will become especially important when studying Euler matrices and transformations between the aircraft *body*, *navigation*, and *inertial* frames.

Rule for EX28: when building a transformation that combines direction vectors using coefficients, **each direction vector is a column**. If the transformation is formulated in the reverse direction, the meanings of the inputs and outputs must also change, and the corresponding inverse matrix must be used.